

# Solving Quadratic Equations by Completing the Square (Part I)

Name  
Date  
Course

## Review

### Equations with Squares

Solve.

Example I)  $-2x^2 + 3 = -5$       Example II)  $-4 = 4 - \frac{(x-5)^2}{2}$

$$\text{Example I) } -2x^2 + 3 = -5$$

$$-2x^2 = -8$$

$$\frac{-2x^2}{-2} = \frac{-8}{-2}$$

$$x^2 = 4$$

$$\sqrt{x^2} = \pm\sqrt{4}$$

$$\boxed{x = \pm 2}$$

$$-8 = -\frac{(x-5)^2}{2}$$

$$-8 = \frac{(x-5)^2}{-2}$$

$$16 = (x-5)^2$$

$$\pm\sqrt{16} = \sqrt{(x-5)^2}$$

$$\pm 4 = x - 5$$

$$5 \pm 4 = x$$

$$\boxed{x = 9 \text{ or } x = 1}$$

Example III)  $3(x+4)^2 - 2 = 73$

$$(x+4)^2 = 25$$

$$\sqrt{(x+4)^2} = \pm\sqrt{25}$$

$$x+4 = \pm 5$$

$$x = -4 \pm 5$$

$$\boxed{x = 1 \text{ or } x = -9}$$

Example IV)  $1 = \frac{x^2}{3} - 2$       Example V)  $-91 = -2(x+3)^2 + 7$

$$9 = x^2$$

$$\pm\sqrt{9} = \sqrt{x^2}$$

$$\boxed{\pm 3 = x}$$

$$49 = (x+3)^2$$

$$\pm\sqrt{49} = \sqrt{(x+3)^2}$$

$$\pm 7 = x+3$$

$$-3 \pm 7 = x$$

$$\boxed{x = 4 \text{ or } x = -10}$$

Example VI)  $\frac{(x-1)^2}{4} - 3 = 6$

$$(x-1)^2 = 36$$

$$\sqrt{(x-1)^2} = \pm\sqrt{36}$$

$$x-1 = \pm 6$$

$$x = \pm 6 + 1$$

$$\boxed{x = 7 \text{ or } x = -5}$$



## Factoring Quadratic Expressions

Factor.

Example VII)  $x^2 + 6x + 8$       Example VIII)  $x^2 + 7x - 18$

$$(x+2)(x+4)$$

$$(x+9)(x-2)$$

Example IX)  $x^2 - x - 2$       Example X)  $x^2 - 13x + 40$

$$(x-2)(x+1)$$

$$(x-8)(x-5)$$

Example XI)  $x^2 + 7x + 10$       Example XII)  $x^2 - 2x - 15$

$$(x+5)(x+2)$$

$$(x-5)(x+3)$$

Example XIII)  $x^2 + 8x - 9$       Example XIV)  $x^2 - 10x + 25$

$$(x+9)(x-1)$$

$$(x-5)(x-5)$$

## Solving Quadratic Equations by Factoring

Solve.

Example XV)  $x^2 + 5x + 6 = 0$       Example XVI)  $0 = x^2 - 7x + 12$

$$(x+2)(x+3) = 0$$

$$x = -2 \text{ or } x = -3$$

$$0 = (x-3)(x-4)$$

$$x = 3 \text{ or } x = 4$$

Example XVII)  $x^2 + 8x - 20 = 0$       Example XVIII)  $0 = x^2 + 11x + 24$

$$(x+10)(x-2) = 0$$

$$x = -10 \text{ or } x = 2$$

$$0 = (x+8)(x+3)$$

$$x = -8 \text{ or } x = -3$$

Example XIX)  $x^2 - 2x - 35 = 0$       Example XX)  $0 = x^2 - 5x + 4$

$$(x-7)(x+5) = 0$$

$$x = 7 \text{ or } x = -5$$

$$0 = (x-4)(x-1)$$

$$x = 4 \text{ or } x = 1$$

Example XXI)  $2x^2 - 5x - 12 = 0$       Example XXII)  $0 = 3x^2 - x - 10$

$$(2x+3)(x-4) = 0$$

$$2x+3=0 \text{ or } x=4$$

$$2x = -3$$

$$x = -\frac{3}{2} \text{ or } x = 4$$

$$0 = (3x+5)(x-2)$$

$$3x+5=0 \text{ or } x=2$$

$$3x = -5$$

$$x = -\frac{5}{3} \text{ or } x = 2$$

Example XXIII)  $5x^2 + 4x - 1 = 0$       Example XXIV)  $0 = 4x^2 - 15x + 9$

$$(5x-1)(x+1) = 0$$

$$5x-1=0 \text{ or } x=-1$$

$$5x = 1$$

$$x = \frac{1}{5} \text{ or } x = -1$$

$$0 = (4x-3)(x-3)$$

$$4x-3=0 \text{ or } x=3$$

$$x = \frac{3}{4} \text{ or } x = 3$$



## Solving Quadratic Equations by Completing the Square

Solve by completing the square.

$$\textcircled{1} x^2 + 8x + 15 = 0$$

$$x^2 + 8x = -15$$

$$x^2 + 8x + 16 = -15 + 16$$

$$(x+4)(x+4) = 1$$

$$(x+4)^2 = 1$$

$$\sqrt{(x+4)^2} = \pm\sqrt{1}$$

$$x+4 = \pm 1$$

$$x = -4 \pm 1$$

$$\boxed{x = -3 \text{ or } x = -5}$$

$$\textcircled{3} x^2 - 12x + 32 = 0$$

$$x^2 - 12x = -32$$

$$x^2 - 12x + 36 = -32 + 36$$

$$(x-6)(x-6) = 4$$

$$(x-6)^2 = 4$$

$$\sqrt{(x-6)^2} = \pm\sqrt{4}$$

$$x-6 = \pm 2$$

$$x = 6 \pm 2$$

$$\boxed{x = 8 \text{ or } x = 4}$$

$$\textcircled{2} 0 = x^2 - 10x + 21$$

$$-21 = x^2 - 10x$$

$$25 - 21 = x^2 - 10x + 25$$

$$4 = (x-5)(x-5)$$

$$4 = (x-5)^2$$

$$\pm\sqrt{4} = \sqrt{(x-5)^2}$$

$$\pm 2 = x-5$$

$$5 \pm 2 = x$$

$$\boxed{x = 7 \text{ or } x = 3}$$

$$\textcircled{4} 0 = x^2 + 4x - 45$$

$$45 = x^2 + 4x$$

$$4 + 45 = x^2 + 4x + 4$$

$$49 = (x+2)(x+2)$$

$$49 = (x+2)^2$$

$$\pm\sqrt{49} = \sqrt{(x+2)^2}$$

$$\pm 7 = x+2$$

$$-2 \pm 7 = x$$

$$\boxed{x = 5 \text{ or } x = -9}$$

$$\textcircled{5} x^2 + 6x + 8 = 0$$

$$x^2 + 6x = -8$$

$$x^2 + 6x + 9 = -8 + 9$$

$$(x+3)(x+3) = 1$$

$$(x+3)^2 = 1$$

$$\sqrt{(x+3)^2} = \pm\sqrt{1}$$

$$x+3 = \pm 1$$

$$x = -3 \pm 1$$

$$\boxed{x = -2 \text{ or } x = -4}$$

$$\textcircled{6} 0 = x^2 - 14x + 40$$

$$-40 = x^2 - 14x$$

$$49 - 40 = x^2 - 14x + 49$$

$$9 = (x-7)(x-7)$$

$$9 = (x-7)^2$$

$$\pm\sqrt{9} = \sqrt{(x-7)^2}$$

$$\pm 3 = x-7$$

$$7 \pm 3 = x$$

$$\boxed{x = 10 \text{ or } x = 4}$$



$$\textcircled{7} x^2 - 16x - 80 = 0$$

$$x^2 - 16x = 80$$

$$x^2 - 16x + 64 = 80 + 64$$

$$(x-8)(x-8) = 144$$

$$(x-8)^2 = 144$$

$$\sqrt{(x-8)^2} = \pm\sqrt{144}$$

$$x-8 = \pm 12$$

$$x = 8 \pm 12$$

$$\boxed{x = 20 \text{ or } x = -4}$$

$$\textcircled{8} 0 = x^2 + 20x + 99$$

$$-99 = x^2 + 20x$$

$$100 - 99 = x^2 + 20x + 100$$

$$1 = (x+10)(x+10)$$

$$1 = (x+10)^2$$

$$\pm\sqrt{1} = \sqrt{(x+10)^2}$$

$$\pm 1 = x+10$$

$$-10 \pm 1 = x$$

$$\boxed{x = -9 \text{ or } x = -11}$$

$$\textcircled{9} 2x^2 + 12x + 16 = 0$$

$$\frac{2x^2 + 12x + 16}{2} = \frac{0}{2}$$

$$x^2 + 6x + 8 = 0$$

$$x^2 + 6x = -8$$

$$x^2 + 6x + 9 = -8 + 9$$

$$(x+3)(x+3) = 1$$

$$(x+3)^2 = 1$$

$$\sqrt{(x+3)^2} = \pm\sqrt{1}$$

$$x+3 = \pm 1$$

$$x = -3 \pm 1$$

$$\boxed{x = -2 \text{ or } x = -4}$$

$$\textcircled{10} 0 = -3x^2 + 30x - 48$$

$$\frac{0}{-3} = \frac{-3x^2 + 30x - 48}{-3}$$

$$0 = x^2 - 10x + 16$$

$$-16 = x^2 - 10x$$

$$25 - 16 = x^2 - 10x + 25$$

$$9 = (x-5)(x-5)$$

$$9 = (x-5)^2$$

$$\pm\sqrt{9} = \sqrt{(x-5)^2}$$

$$\pm 3 = x-5$$

$$5 \pm 3 = x$$

$$\boxed{x = 8 \text{ or } x = 2}$$

$$\textcircled{12} 0 = -5x^2 - 20x + 25$$

$$\frac{0}{-5} = \frac{-5x^2 - 20x + 25}{-5}$$

$$0 = x^2 + 4x - 5$$

$$5 = x^2 + 4x$$

$$4 + 5 = x^2 + 4x + 4$$

$$\textcircled{11} 4x^2 - 32x + 28 = 0$$

$$\frac{4x^2 - 32x + 28}{4} = \frac{0}{4}$$

$$x^2 - 8x + 7 = 0$$

$$x^2 - 8x = -7$$

$$x^2 - 8x + 16 = -7 + 16$$

$$(x-4)(x-4) = 9$$

$$(x-4)^2 = 9$$

$$\sqrt{(x-4)^2} = \pm\sqrt{9}$$

$$x-4 = \pm 3$$

$$x = 4 \pm 3$$

$$\boxed{x = 7 \text{ or } x = 1}$$

$$9 = (x+2)(x+2)$$

$$9 = (x+2)^2$$

$$\pm\sqrt{9} = \sqrt{(x+2)^2}$$

$$\pm 3 = x+2$$

$$-2 \pm 3 = x$$

$$\boxed{x = 1 \text{ or } x = -5}$$